SEMINAR PRESENTATION

A Computational Investigation on Hydrogen Embrittlement in Pipeline Steels

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Outline

- Introduction
- Literature Review
- Research Gap
- Objectives
- Methodology and Results
- Summary and Future Work

Introduction

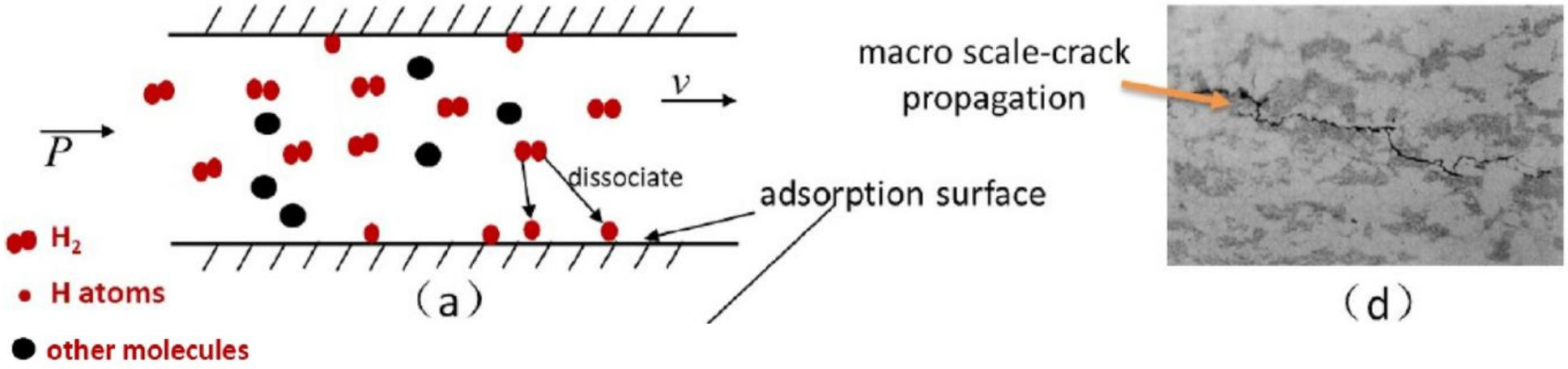
The Problem: H Embrittlement

A hidden threat in metals

- Atomic H enters steel from welding, corrosion, H_2 gas, etc.
- Even small amounts rapidly reduce ductility: sudden failure below design stress

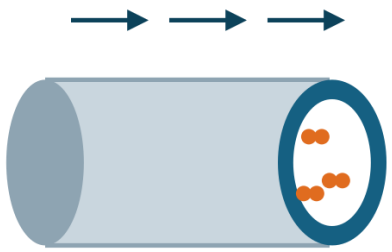
Why look at Pipeline Steels?

- Gas grids are being retrofitted for H_2: 5-10x cheaper than new lines
- Lines with welds and dents as stress raisers
- Sour (H_2S) service causes HIC, blistering

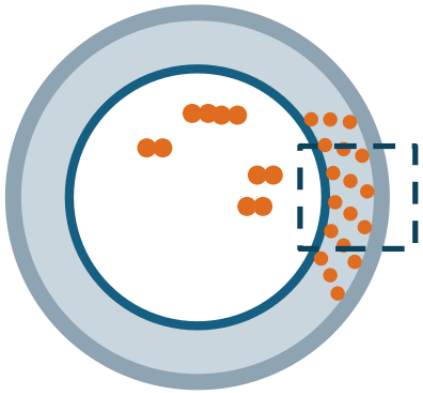


HE Mechanism in Pipeline Steel

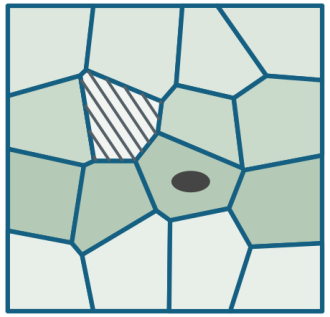
- **Pipeline Steels: A Critical Structural Material**
  - High pressure, long distance transportation of hydrogen and natural gas
  - High strength + toughness requirements
  - Complex multiphase microstructure
  - Hydrogen ingress and diffusion through the steel
  - Hydrogen embrittlement



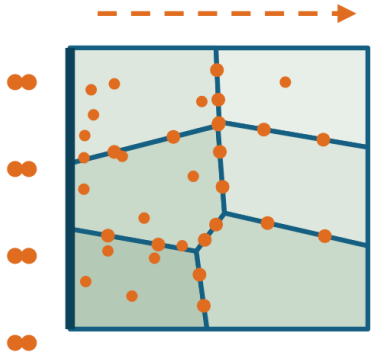
Pipeline
X52-X80 grade line pipe carrying H_2 /sour gas



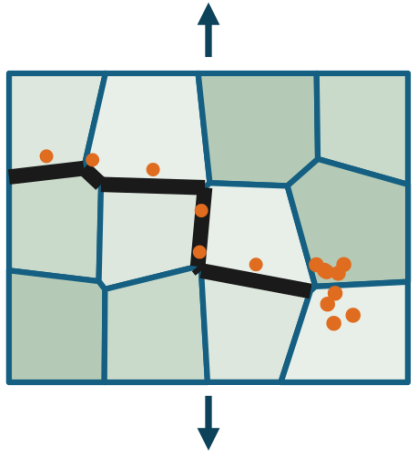
Cross sectional view showing atomic H entering through inner wall



Steel microstructure depicting ferrite grains, pearlite and inclusion

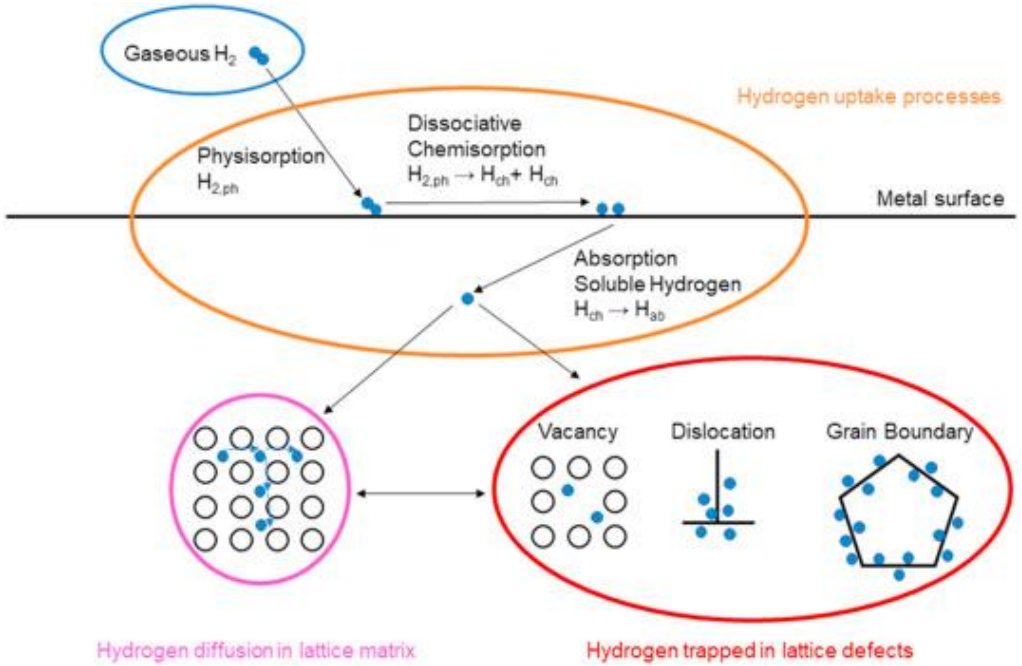


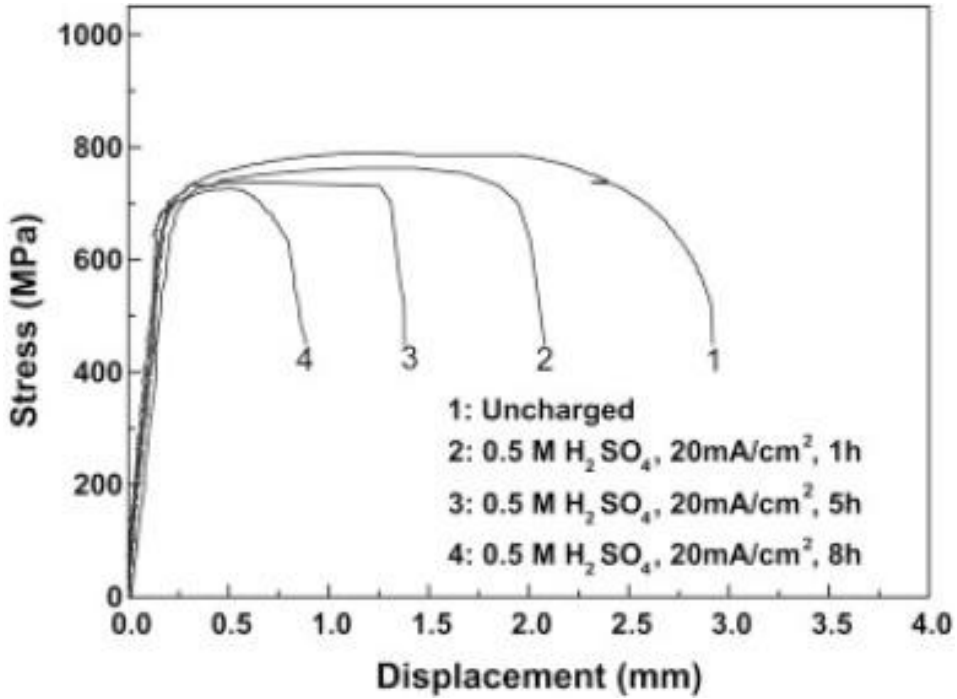
H uptake and diffusion
H_2 -> 2H at surface
H moves through lattice and along GBs



H builds up at defect sites under stress and macro cracks form

H Uptake Process and Charging Effects





- H_2 physisorbs, dissociates into atomic H at metal surface and is absorbed into the lattice
- Dissolved H diffuses and gets trapped at vacancies, dislocations and grain boundaries

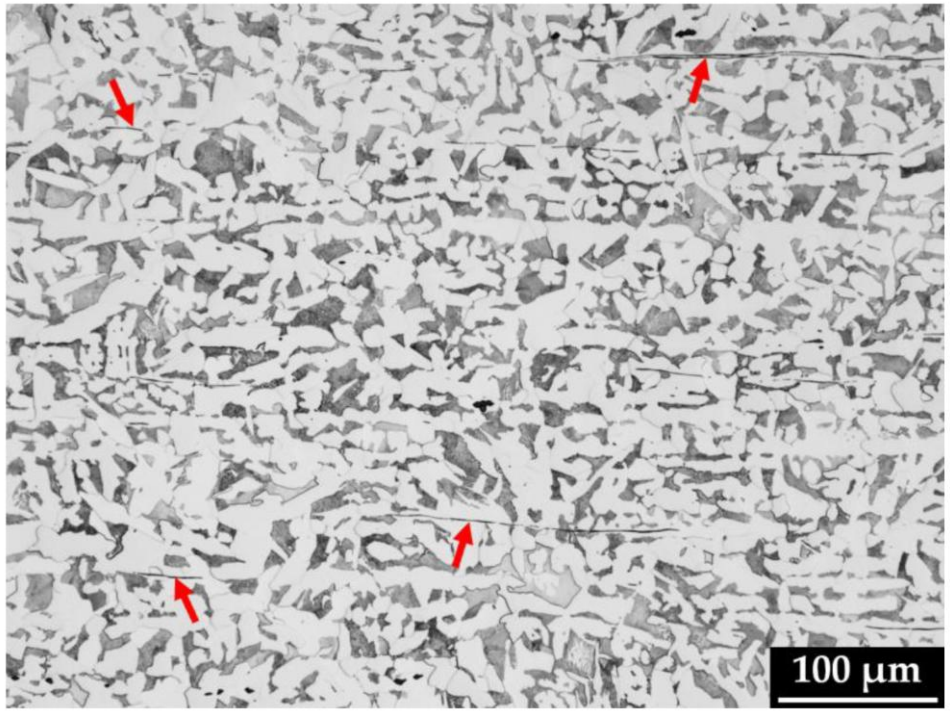
- Stress-strain curves of a X100 pipeline steel after various durations of hydrogen charging
- Longer charging leads to more embrittlement; H costs ductility, not strength

Michler and Naumann, 2010; Pourazizi et al., 2020a
Dong et al., 2009

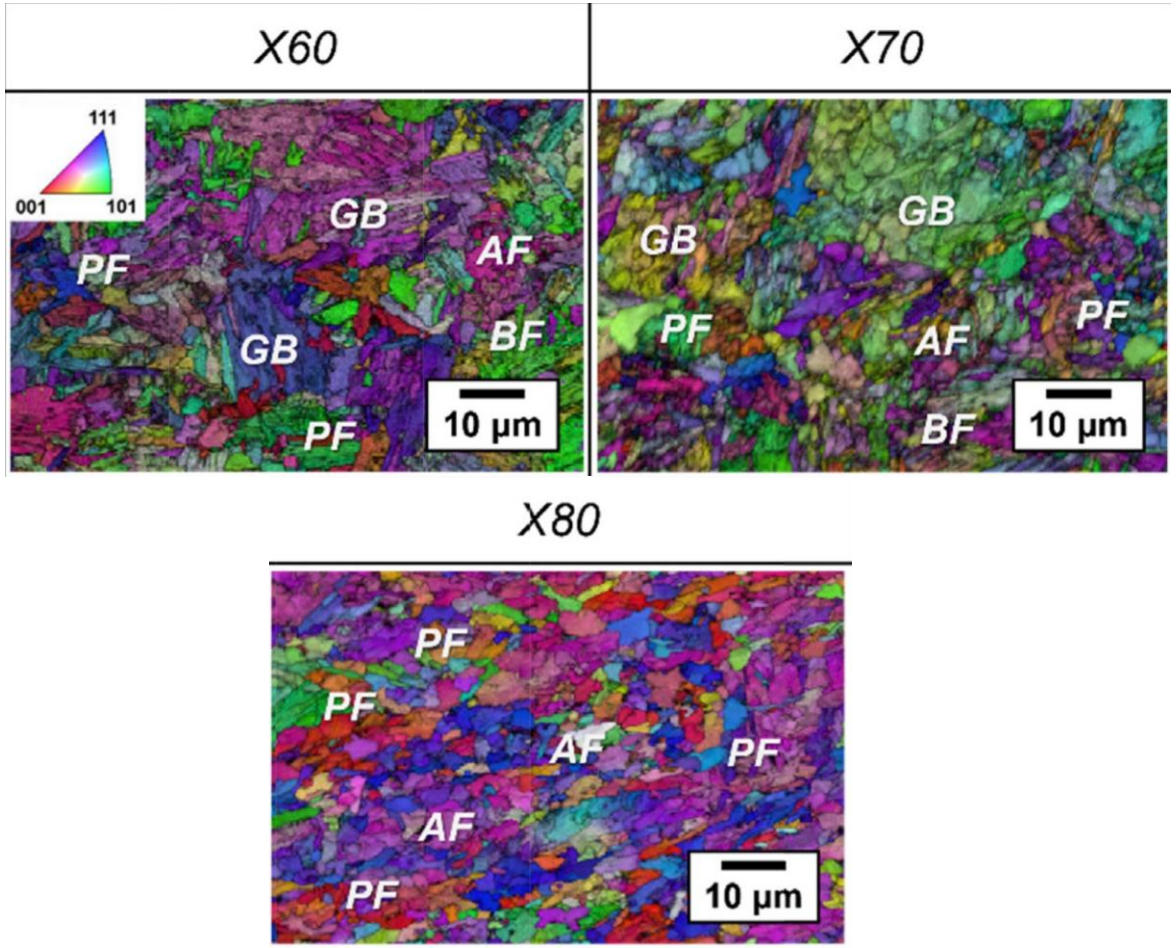
Pipeline Steel Grades Based on YS and Phases

APL 5L Grade	Minimum Yield Strength (MPa)	Phase Constituents
Grade B , X42-X52	Grade B: ~245 MPa X42-X52: ~ 290-360 MPa	Polygonal ferrite and pearlite
X60-X70	~415-485 MPa	Quasi-polygonal ferrite, granular bainite and acicular ferrite
X80	~555 MPa	Fine acicular ferrite, lower bainite, minor M/A constituents
X100	~690 MPa	Granular and lath bainite, hard M/A constituents, nanoscale ppts (Ti, Nb)(C,N)

Microstructures of Various Pipeline Steels



Ferrite-pearlite microstructure in a section of base metal in X46 pipe steel

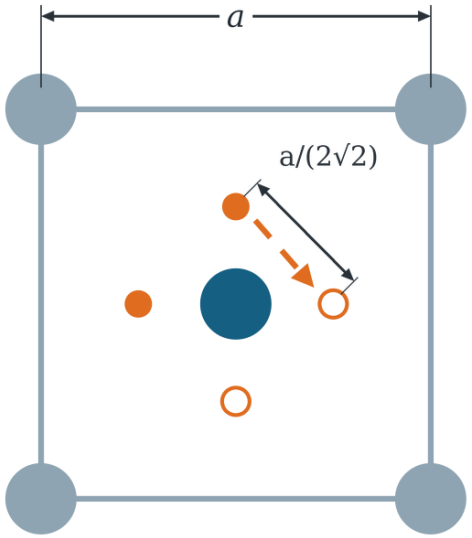


EBSD maps of API X60, X70 and X80 pipeline steels

Hydrogen Diffusion and Trapping

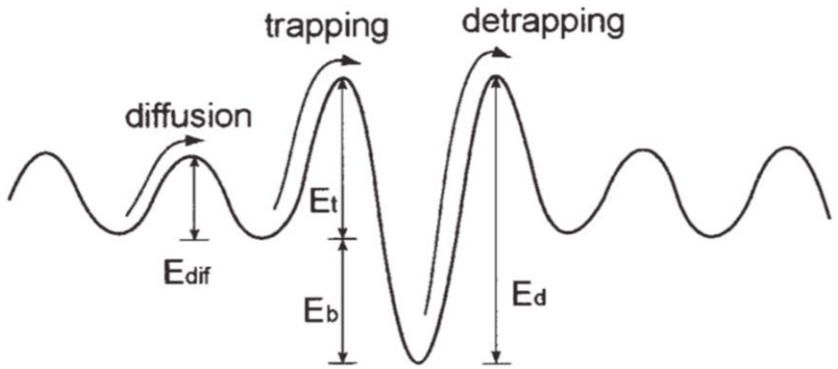
Lattice Diffusion

- **Fick's Law**: $J = -D_L \nabla C_L$
- **Mass balance**: $\partial C_L/\partial t = -\nabla \cdot J \Rightarrow \partial C_L/\partial t = D_L \nabla^2 C_L$
- H hops between tetrahedral sites of BCC Fe ($D_L \approx 7\times10^{-9}$ m^2/s at room temperature)



Trapping

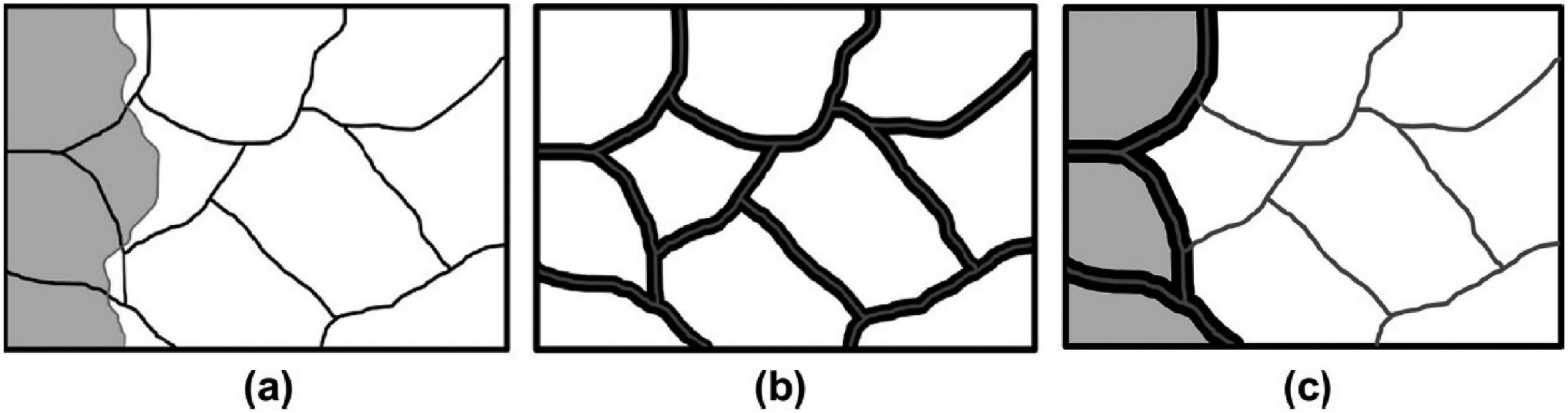
- Total H: $C = C_L + C_T$ (lattice + trapped)
- **Oriani (local equilibrium):** $D_{app} = D_L / [1 + (N_T/N_L) \exp(E_b/RT)]$
- **Reversible** (E_b < 35 kJ/mol): dislocations, GBs, interfaces
- **Irreversible** (E_b > 35 kJ/mol): carbides, inclusions



E_{dif}: diffusion energy
E_t: trapping activation energy
E_b: trap site binding energy
E_d: detrapping activation energy

Source: Wei et al., Metall. Mater. Trans. B, (2004)

H Diffusion Paths in Polycrystals

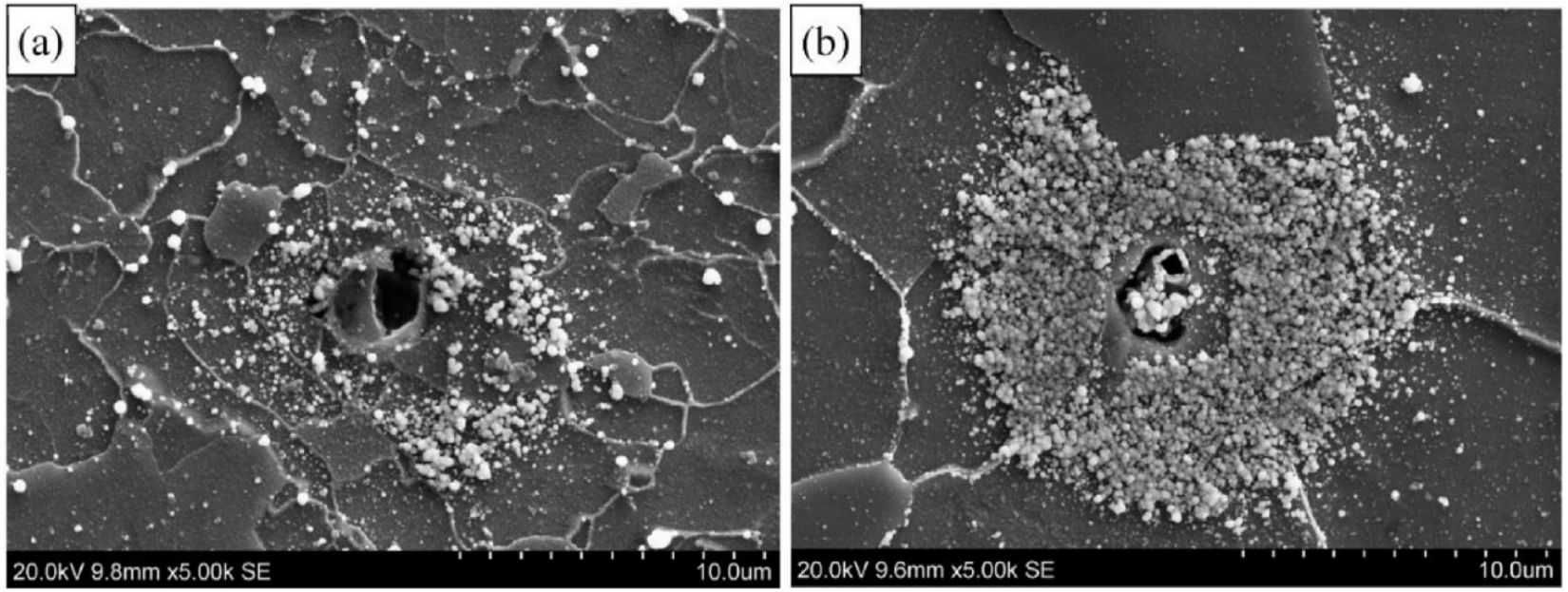


Three scenarios for the hydrogen diffusion paths:

a) bulk diffusion through the interiors of crystals and grains
b) grain boundary diffusion along the disordered internal interfaces of poly-crystals
c) surface diffusion over single crystals or poly-crystals

Source: Mohtadi-Bonab & Masoumi, JMR&T, 2023

HMT Experiment



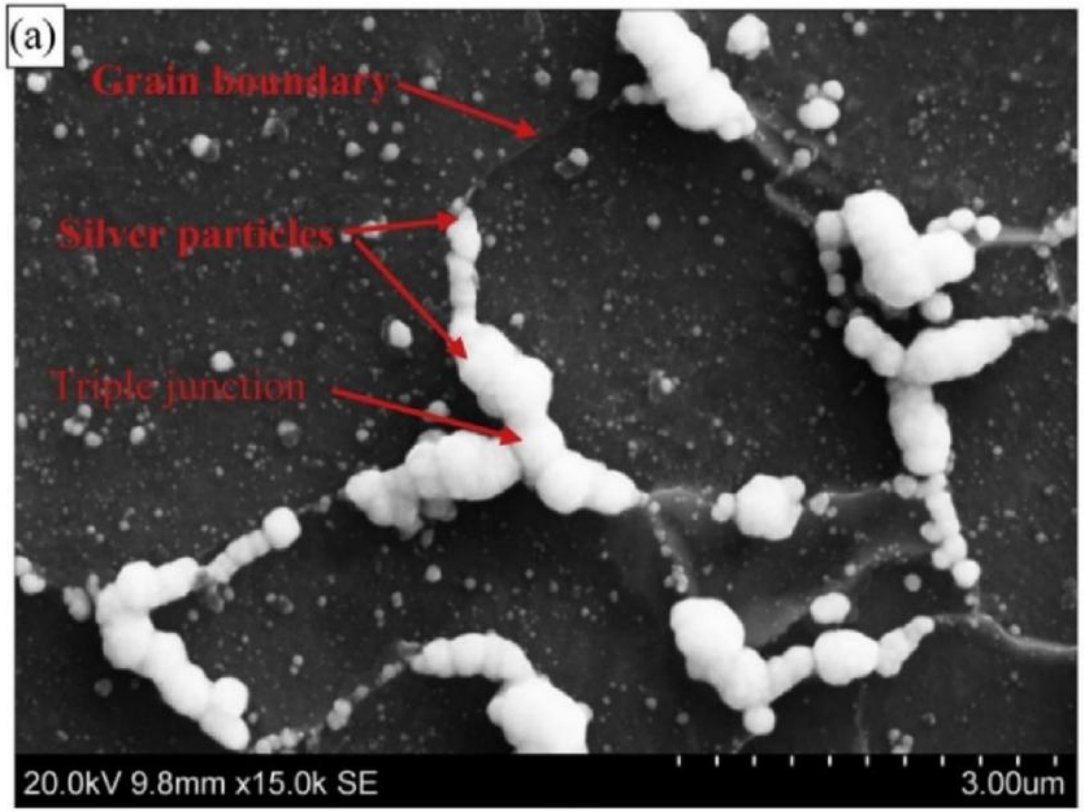
- Escaping H reduces Ag^{+} to Ag, so silver marks where H leaves the steel
- H is trapped at the inclusion–matrix interface
- Mn-based inclusions trap more than Ti-based ones

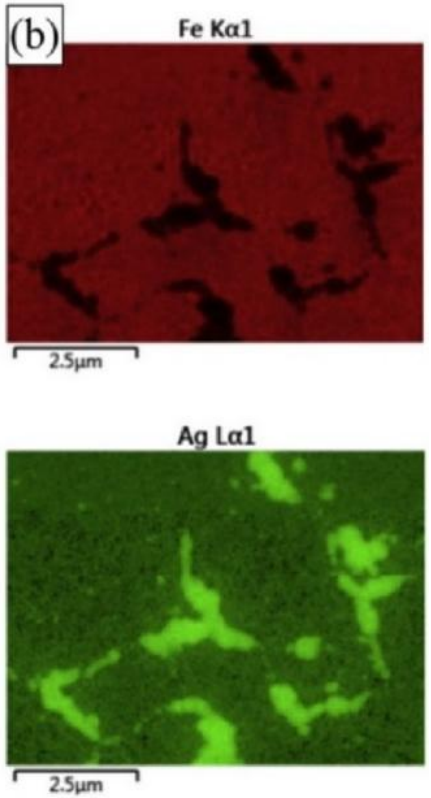
Hydrogen diffusion path observed around (a) Ti based inclusion and (b) Mn based inclusion in X70 pipeline steel

(a)- as received, (b)-annealed

Source: Thomas & Szpunar, Int. J. Hydrogen Energy, 2020

Literature Review





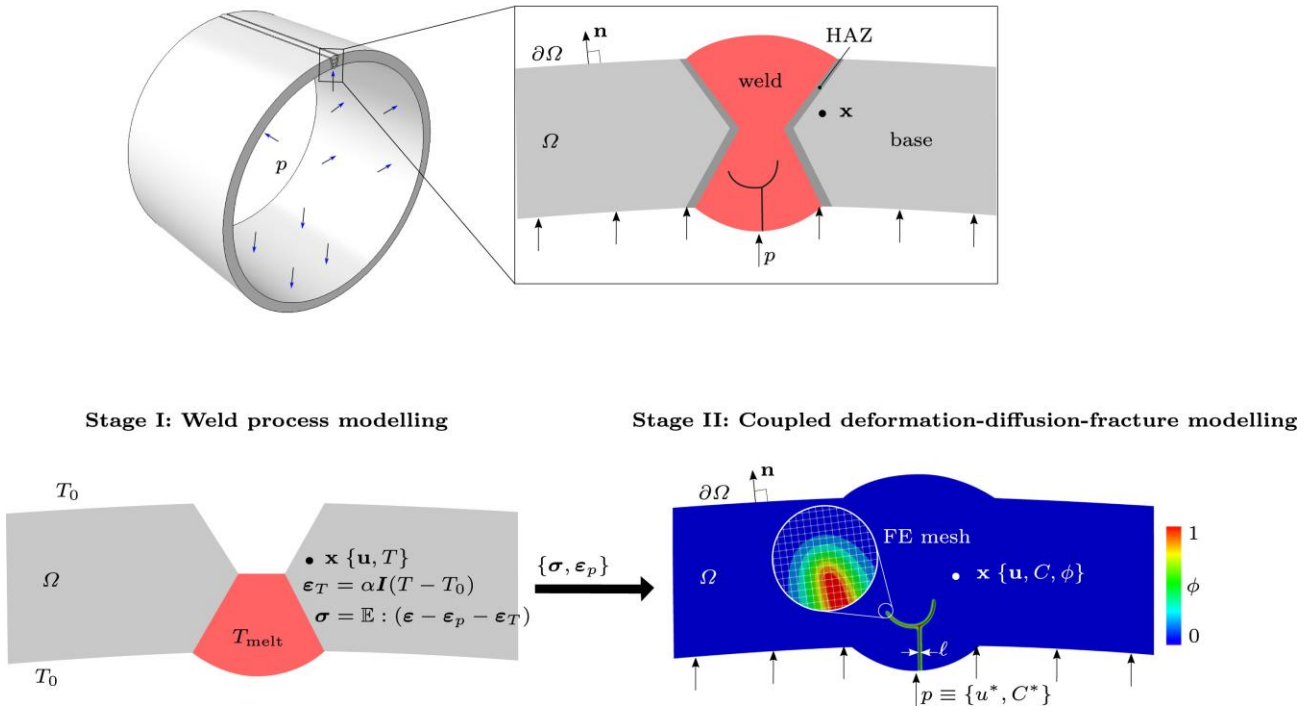
SEM image of hydrogen diffusion path at the triple junctions in X70 pipeline steel (as-received case)

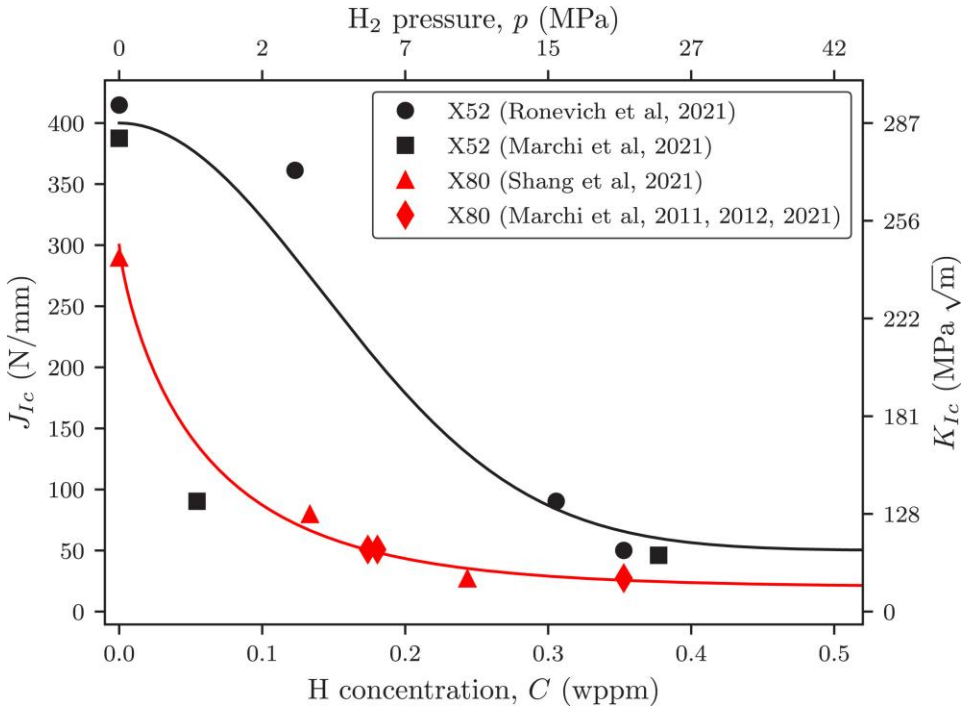
- Silver lines the boundaries and is densest at triple junctions, so these are the fastest H paths
- H transport is microstructure-controlled.
  Coarser grains mean fewer GB and triple-junction paths and less trapping.

Source: Thomas & Szpunar, Int. J. Hydrogen Energy, 2020

Literature Review

Phase-Field Modelling of Welded H_2 Pipelines





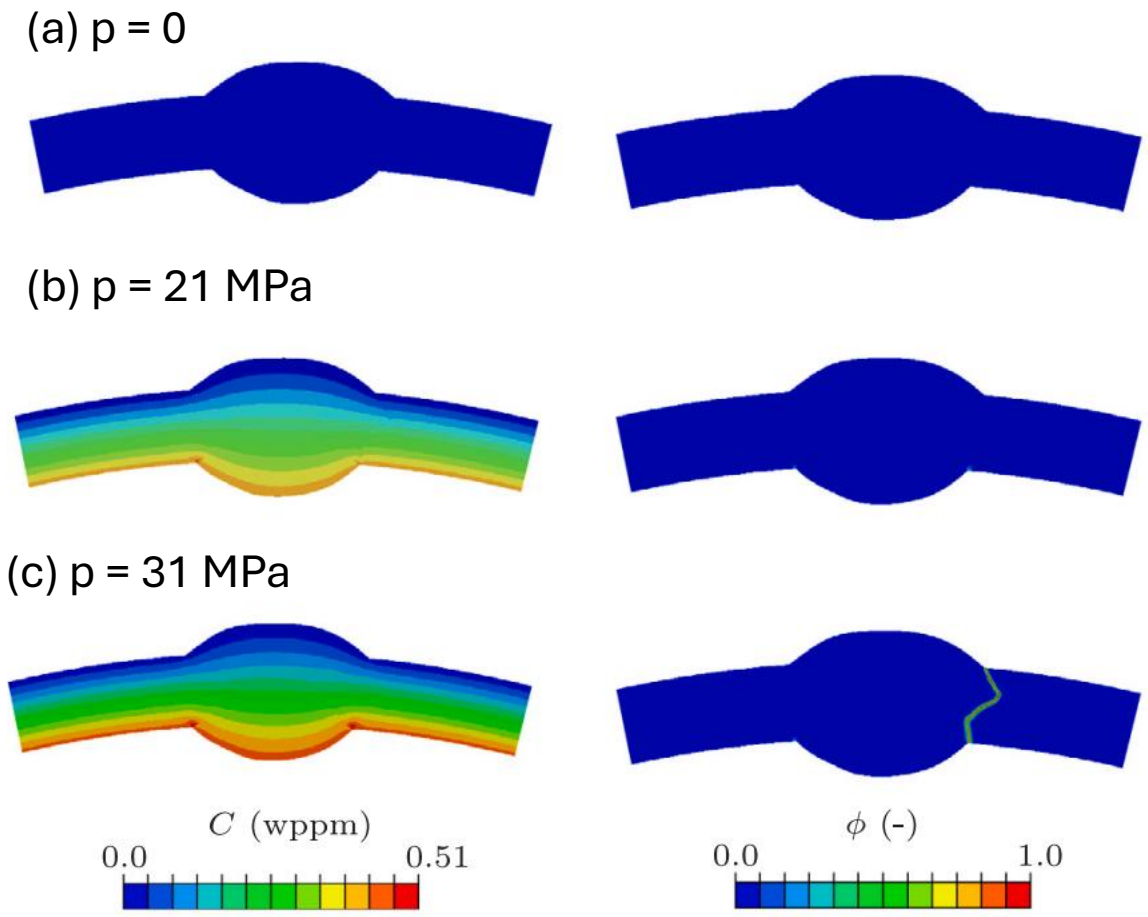
Stage I: Weld simulation gives residual stress, plastic strain
Stage II: Phase-field fracture with H-dependent toughness

Toughness of X52 and X80 collapses with ~0.1-0.3 wppm H

Source: Martínez Pañeda et al., Int. J. Hydrogen Energy (2025)

Literature Review

Key Predictions for Seam Welds



Results obtained for two-pass weld considering X80 pipeline steel as the base and weld materials

- X80 weld cracks once residual stresses are included
  (yields at ~45 MPa without them)
- Accounting for weld microstructure cuts the failure pressure, but its effect on H diffusion is *assumed* from Vickers hardness data.

Predictions for lattice H concentration (C) and phase field fracture (ϕ) as a function of pipeline pressure.

Source: Martínez Pañeda et al., Int. J. Hydrogen Energy (2025)

Research Gap

- Macro pipeline models use assumed or hardness-based diffusivity, with traps lumped into an effective diffusivity
- Microstructure-resolved diffusion models exist, but are rarely applied to pipeline-steel microstructures
- Experiments (HMT, permeation) show that microstructure controls H paths and trapping, but they can't isolate grain size, phase and inclusions from one another
- No well-established link from microstructure → Deff → component-scale prediction

Objectives

- Simulate H diffusion in equiaxed ferrite RVEs with explicit grain boundaries (PHIMATS)
- Quantify the effect of grain size and morphology on D_{eff} and GB segregation
- Extend to acicular and lamellar ferrite, relating D_{eff} to C content
- To study diffusion for the different microstructures of X52-X100 pipe steel grades
- Feed microstructure-derived D_{eff} into a macro-scale pipeline model and compare it with assumed D

Methodology

Mesoscale RVE-FEM Framework (PHIMATS)

Microstructure
Polydisperse grain map

→

Phase-field GBs
Order parameters ϕ_i mark grains and GBs

→

FE Model
Mesh, BCs, material data

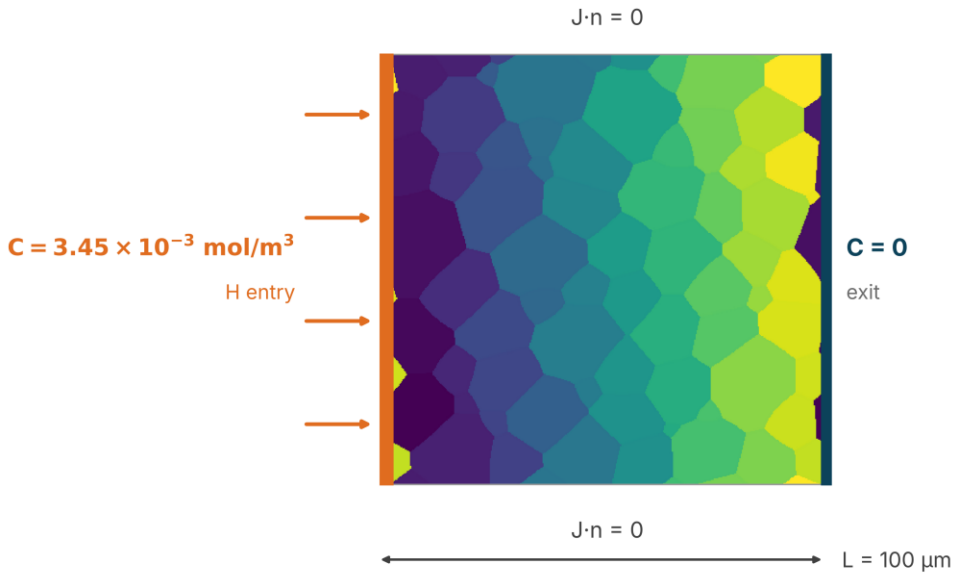
→

Solve
Diffusion + GB trapping, time stepped

→

Upscale
D_{eff} -> macro pipeline model

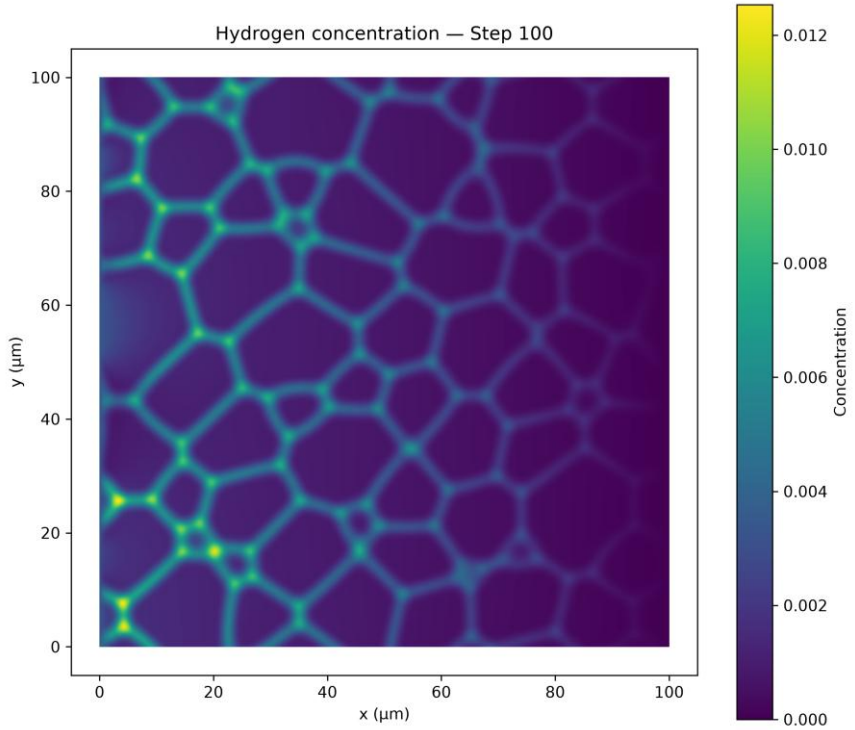
RVE and Boundary Conditions



- 2D RVE, 100 × 100 µm, polydisperse equiaxed ferrite
- GBs as diffuse phase-field regions that trap H
- Flux = Fickian diffusion + GB trapping term
- Permeation case: H enters at x = 0, leaves at x = L
- Outputs: C, |J| maps and $D_{eff} = J_{ss} \cdot L/\Delta C$ (J_{ss}: steady state flux)

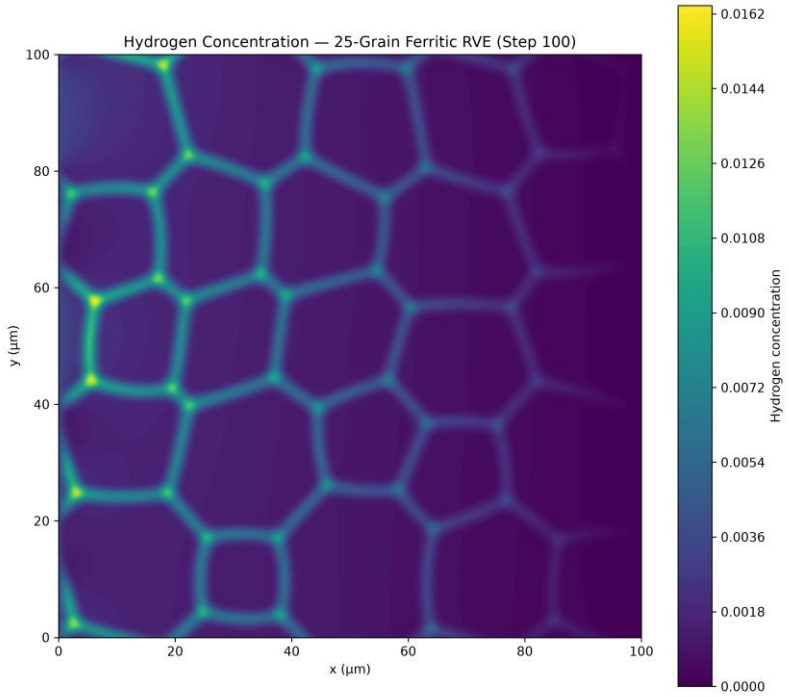
Results

Polydisperse Equiaxed Ferrite RVE



(a) H concentration (fine case)

H concentrates along GBs, peaking at triple junctions



(b) H concentration (coarse case)

|J| along GBs is higher than in grain interiors

$D_{eff} = J_{ss} \cdot L/(C_L - C_R)$

$J_{ss} = 3.155 \times 10^{-7}$ m/s

L=100 μm, ΔC=0.003453

$D_{eff,\,a} \approx 9.137 \times 10^{-9}\ m^2/s$

$D_{eff,\,b} \approx 9.804 \times 10^{-9}\ m^2/s$

(using TrapGB model parameters)

C and |J| are highest near x = 0 and decay towards the exit

Summary and Future Work

- H transport in pipeline steels is microstructure-controlled; macro models still assume D
- Built a PHIMATS RVE–FEM framework with explicit GBs for equiaxed ferrite
- GBs act as H-enriched, high-flux paths, with peaks at triple junctions

Future Work

- Decrease grain size and extend to acicular ferrite and pearlite, other phases like bainite, martensite and retained austenite of X60-X100 grades
- Add stress-assisted diffusion and strain-induced trapping
- Feed microstructure-derived D_{eff} into a macro pipeline model

Acknowledgement

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- Department of Materials Science and Engineering, IIT Kanpur - for providing the necessary research facilities and support
- Lab members and colleagues - for their assistance, valuable discussions and cooperation
- Family and friends - for their constant encouragement and support



STEGAL natural gas pipeline construction site, Germany